Literature status: Kato's positive-commutator conjecture is false

Answer-identification packet

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Status: literature already answered (negative). The exact Kato conjecture formulated as Conjecture 1.4 in arXiv:1804.05227 was explicitly disproved by Frank and Ivanisvili, arXiv:2608.07805.

Original question

I. Herbst and T. L. Kriete, *The Howland–Kato Commutator Problem*, arXiv:1804.05227 (2018; published as a 2019 book chapter), Conjecture 1.4 on page 3, formulate:

> If $C = i[f(P), g(Q)] \geq 0$ is nonzero and both bounded real functions f, g are increasing, must there be $a, b > 0$ with $ab = \pi/2$, $f \in K_a$, and $g \in K_b$?

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For convenience let us formulate a conjecture which we shall call K (after Kato):

Conjecture 1.4. Suppose $C = i[f(P), g(Q)] \geqslant 0$ and in addition $C \neq 0$. Suppose both f and g are increasing. Then there exist a and b with $ab = \pi/2$ such that $f \in K_a$ and $g \in K_b$.

Unfortunately we are far from proving K. But in the rest of this paper we will give several results which illuminate the properties of the set of f and g for which

Explicit later answer

R. L. Frank and P. Ivanisvili, *A counterexample to the Kato conjecture for positive commutators*, arXiv:2608.07805 (7 August 2026), explicitly cite the Herbst–Kriete formulation. Their Theorem 1 proves

$$C = i[\arctan(P), \arctan(Q)] \geq 0, \qquad \operatorname{Tr} C = \frac{\pi}{2},$$

so $C \neq 0$. Their Proposition 15 proves that $\arctan \in K_1$ but $\arctan \notin K_a$ for $a > 1$, while $-\arctan$ is in no K_a with $a > 0$. Thus no correlated sign and strip widths can satisfy $ab = \pi/2 > 1$. Corollary 3 states verbatim that the conjecture fails.

$$\operatorname{Im} h(z) \operatorname{Im} z \geq 0 \qquad \text{for all } z \in S_a\,. \tag{3}$$

Kato [5] proved that

$$f \in K_a, \quad g \in K_b, \quad ab \geq \frac{\pi}{2} \qquad \Longrightarrow \qquad i[\, f(P), g(Q)\,] \geq 0\,. \tag{4}$$

Since restriction to a smaller strip preserves membership in the corresponding Kato class, the condition $ab \geq \pi/2$ can equivalently be written with equality after decreasing one of the strip widths.

The conjectural converse, in the formulation of Herbst and Kriete [3], is the following.

Conjecture 2 (Kato conjecture)**.** *Let f and g be bounded real-valued functions and suppose that*

$$i[\, f(P),\, g(Q)\,] \geq 0, \qquad i[\, f(P),\, g(Q)\,] \neq 0.$$

After replacing both f and g by their negatives if necessary, there are $a, b > 0$ with $ab = \pi/2$ such that

$$f \in K_a, \qquad g \in K_b.$$

More precisely, of course, the conjecture is that there are function $\tilde{f} \in K_a$ and $\tilde{g} \in K_b$, coinciding with f and g almost everywhere.

It is easy to see that the function arctan belongs to K_1, but not to K_a for any $a > 1$, and that the function $-\arctan$ does not belong to K_a for any $a > 0$; see Proposition 15. Therefore, Theorem 1 implies the following.

Corollary 3. *Conjecture 2 fails.*

Let us put Conjecture 2 into its historical context. The problem arose from work of Howland [4] on dense point spectrum. In 1987 he observed that the pair

$$f(x) = \arctan(x/2), \qquad g(x) = \tanh x$$

gives a nonnegative commutator $i[\, f(P), g(Q)\,]$. Kato subsequently began a systematic study of all pairs of bounded real functions producing a positive commutator [5] and proved several interesting results.

Classification and scope

This is an exact separate-paper answer, not an agent-inferred implication: the supporting authors name the Howland–Kato problem, cite Herbst–Kriete, restate the same conjecture, and explicitly declare it false. The conjecture is therefore fully settled negatively. The broader classification problem for all positive commutators remains a separate research programme.

References

[1] I. Herbst and T. L. Kriete, *The Howland–Kato Commutator Problem*, arXiv:1804.05227. https://arxiv.org/abs/1804.05227.

[2] R. L. Frank and P. Ivanisvili, *A counterexample to the Kato conjecture for positive commutators*, arXiv:2608.07805 (2026). https://arxiv.org/abs/2608.07805.